Does Time-Series Transformer Pre-Training Help LSST Classification?

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github.com/elyanrgn/Time-Series-Classification

Deep Learning For Time Series

Outline

1. Introduction & Motivation
2. Methodology: Data & Models
3. Transfer Strategies
4. Results & Analysis
5. Conclusion

Motivation & Research Question

- **Context:** Pre-trained transformers have revolutionized NLP/Vision via transfer learning.
- **Problem:** In Time-Series, the benefit of pre-training remains unclear under **domain shift**.
- **LSST Challenge:** Classification of multivariate astronomical light curves (6 channels), from short windows with high noise and .

Research Question

Does supervised pre-training on industrial data (ETTh1) improve astronomical classification (LSST) compared to training from scratch?

Data & Tasks: Mismatch by Design

Source: ETTh1 (Forecasting)

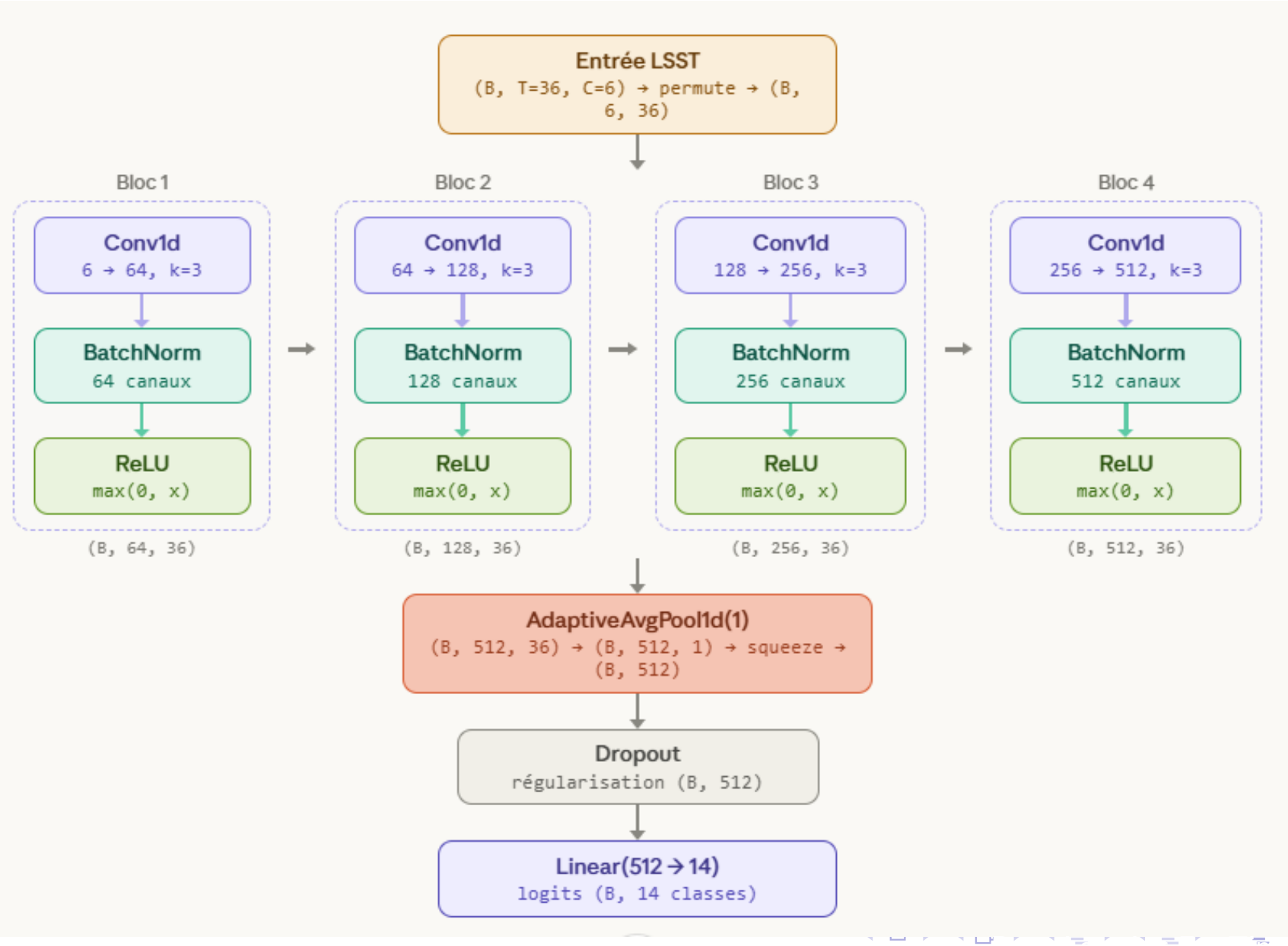
- Industrial temperatures.
- Regression task (predict 24 steps).
- Regular, long sequences.

Target: LSST (Classification)

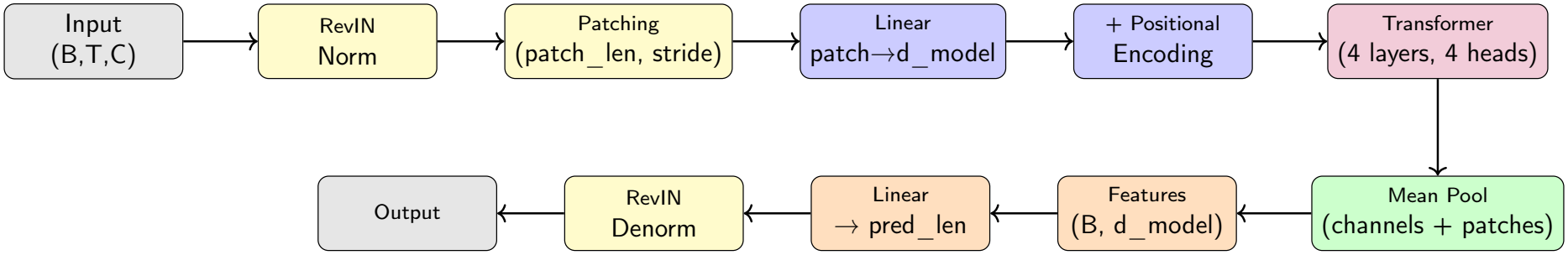
- Astronomical light curves.
- 14-class classification.
- Short (36 steps), noisy, multivariate.

Identified Window Length = 36 for both tasks to ensure identical positional embeddings.

CNN Baseline Architecture



Architecture IndPatchTST



Training

CNN Baseline

- GridSearch to find the best configuration.
- Results averaged over 15 different seeds.

IndPatchTST

- **ETTh1** : Used Optuna to found the best hyperparameters
- **LSST**: Added a classification head and compared 4 training methods. Tuning done with Optuna in all cases.

Four Transfer Strategies

To assess the impact of pre-training, we compare:

1. **From Scratch**: Random initialization (No pre-training).
2. **Head Only**: Backbone frozen; only the MLP head learns.
3. **Late Encoders**: Last layers + head unfrozen.
4. **Full Fine-Tune**: Entire model learns with **Differential LRs** ($\eta_{backbone} < \eta_{head}$).

Main Results: The "From Scratch" Superiority

Table: LSST test performance (Mean $\pm$ Std over 15 runs)

Strategy	**Accuracy**	**F1 (Macro)**
CNN Baseline	0.559 ± 0.021	0.398 ± 0.021
From Scratch (TST)	**0.618** ± 0.010	**0.416** ± 0.028
Pre-trained (ETTh1)		
Full Fine-Tune	0.543 ± 0.014	0.297 ± 0.018
Head Only	0.315 ± 0.000	0.034 ± 0.000

- **Insight 1:** The Transformer architecture outperforms CNN by $\sim$6%.
- **Insight 2:** Supervised pre-training on ETTh1 causes **negative transfer**.

Extension: MOMENT-1-Large

- **Objective:** Test a true Foundation Model (LTM) pre-trained on 1.1 billion points.
- **Strategy:** Full Fine-Tuning ($\sim$30h training).
- **Result:** Accuracy ≈ 0.562.

The Imbalance Wall

- High performance on majority classes (F1 ≈ 0.85).
- **F1 = 0.00** for 5 rare transient classes out of 14.
- Foundation Models don't solve intrinsic class imbalance!

Key Takeaways & Conclusion

- **Architecture > Pre-training:** A well-tuned Transformer from scratch is the most robust approach for LSST.
- **Domain Mismatch:** Pre-training on industrial data is not transferable to astronomical transients.
- **Methodology:** Robustness achieved via **15 seeds** and separate hyperparameter searches (Optuna).

Thank you for your attention!

`GitHub Repository`